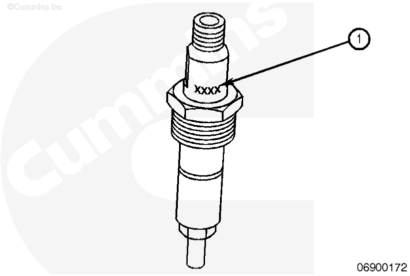
Trainings

General Information

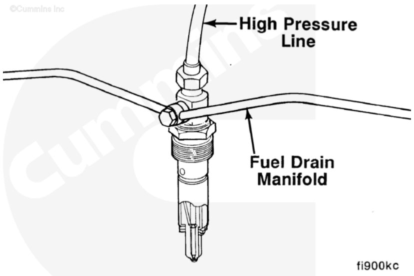
⚠ CAUTION ⚠

Use only the specified injector for the engine.

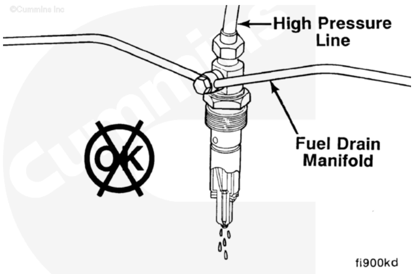
All engines use closed nozzle, hole-type injectors. However, the injectors can have different part numbers for different engine ratings. The last four digits of the Cummins® part number are used to identify the injectors.



During the injection cycle, high pressure from the injection pump rises to the operating (pop) pressure, which causes the needle valve in the injector to lift. Fuel is then injected into the cylinder. A shimmed spring is used to force the needle valve closed as the injection pressure drops below the pop pressure to seal off the nozzle after injection.



Failure of the needle valve to lift and close at the correct time or the needle valve stuck open can cause the engine to misfire and produce low power. Fuel leaking from the open nozzle can cause a fuel knock, poor performance, smoke, poor fuel economy, and rough running.

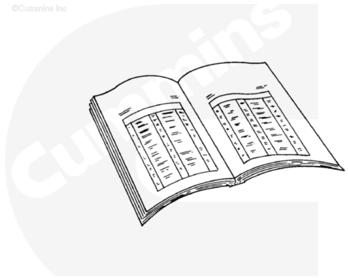


Preparatory Steps

Front Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



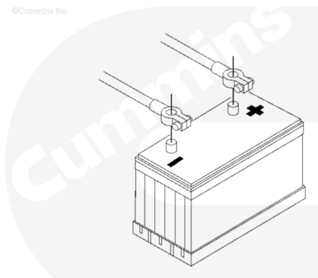
ck800wa

- Disconnect the batteries.
- Clean around the injectors.
- Disconnect the high-pressure fuel supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)
- Disconnect the fuel drain manifold. Refer to Procedure 006-021 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-021-tr.html)

Rear Gear Train

⚠ WARNING ⚠

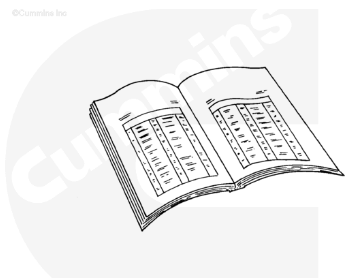
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



13900050

- Disconnect the batteries.

- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Remove the injector supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)
- Remove the fuel connector. Refer to Procedure 006-052 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-052-tr.html)



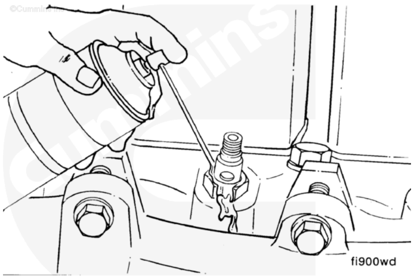
ck800wa

Note : The fuel connector must be removed before removing the injector or damage to the connector will result.

Remove

Front Gear Train

Rust-Penetrating Solvent



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

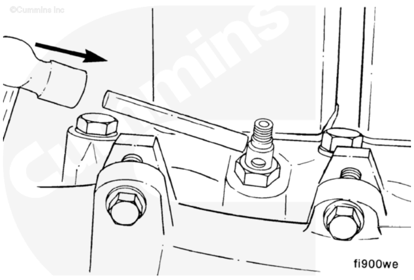
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

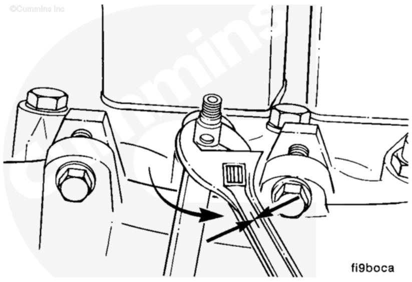
When rust has formed on the hold-down nut, the injector can turn in the bore when the nut is loosened. This can cause severe damage to the head by the injector locating ball cutting a groove in the bore.

Soak the hold-down nut with a rust-penetrating solvent for a minimum of 3 minutes.

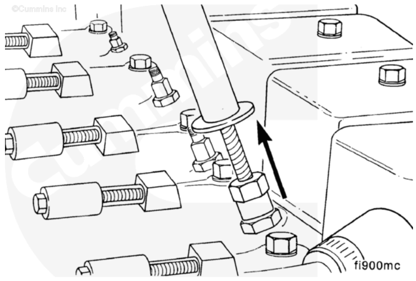
Hit the injector body with a drift pin to loosen any rust.



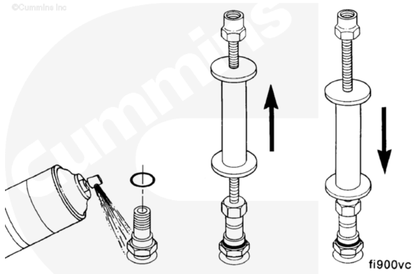
Hold the injector body with an adjustable wrench while loosening the hold-down nut with a 24-mm box wrench.



Use an injector puller, Part Number 3823276, to remove the injectors.



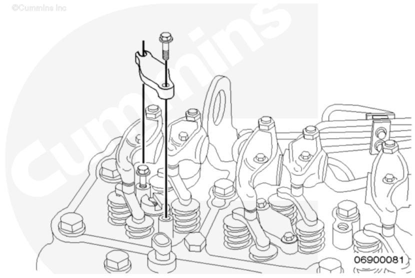
It is often necessary to tap the injector with the injector puller to work the injector up and down to remove it.



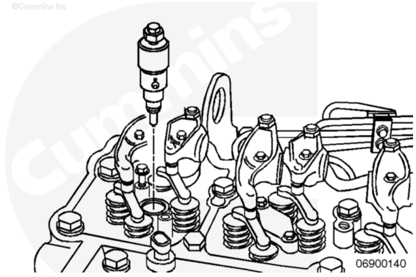
Rear Gear Train

Remove the injector hold-down capscrews.

Tilt the hold-downs up, and slide them out.



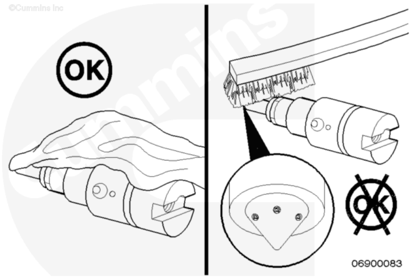
Use an injector puller, Part Number 3825156, to remove the injectors from the head.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



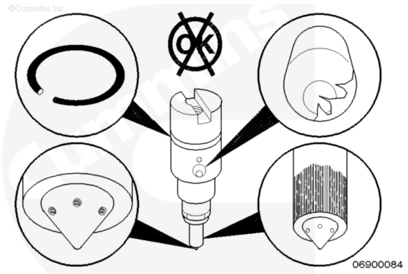
⚠ CAUTION ⚠

Do not use any abrasives (such as glass beading, sand paper, emery cloth, Scotch-Brite™ pads, etc) or metallic items (including wire brushes made of any metallic material) to clean the injectors. The use of any cleaning method other than safety solvent and a soft, clean, lint-free cloth will damage the nozzle holes and cause performance issues.

Clean the injector tip and body with safety solvent and a soft, clean rag.

Note : If necessary, use a **brass** brush to clean off carbon.

Inspect the injector for burrs on the inlet to the injector.



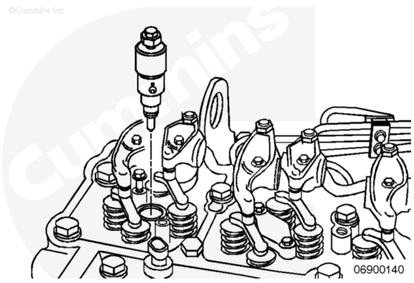
Inspect the nozzle holes for any signs of damage, such as erosion or plugging.

Inspect the nozzle color for signs of overheating.

Note : Overheating will cause the nozzle to turn a dark yellow/tan or blue color, depending on the degree of overheating.

Inspect the o-ring for damage.

O-rings and Sealing Washers



If the high-pressure connectors are **not** damaged, remove the injectors and inspect the o-rings and sealing washers. If there is **any** damage to the injector parts, replace the o-ring or sealing washers.

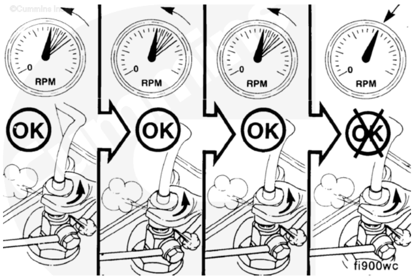
Inspect the contact point of the sealing washers to the cylinder head.

Test

Front Gear Train

⚠ WARNING ⚠

Do not vent the fuel system on a hot engine; this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.



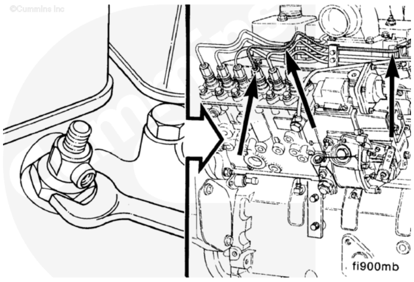
To determine which cylinder is misfiring, operate the engine, loosen the fuel line nut at one injector, and listen for a change in engine speed.

Note : A drop in engine speed indicates the injector was delivering fuel to the cylinder.

Note : Be sure to tighten the fuel line nut before proceeding to the next injector.

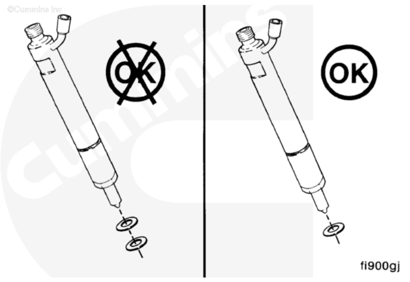
Check each cylinder until the malfunctioning injector is found.

Remove the malfunctioning injector to test or replace it.

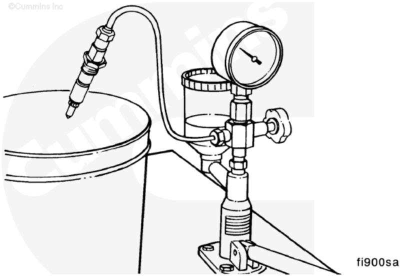


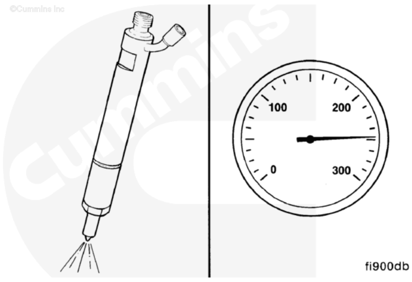
If the engine continues to misfire after replacing the injector, check for leaks in the high-pressure line. Also, check for a damaged delivery valve that lets the fuel drain back into the injection pump.

Check for an extra copper sealing washer on the injector.



Carbon buildup in the orifices in the nozzle will also cause low power from the engine. Remove and check the spray pattern, or replace the injectors.

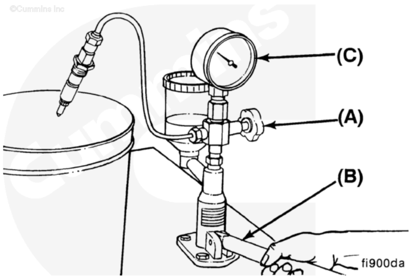




⚠ WARNING ⚠

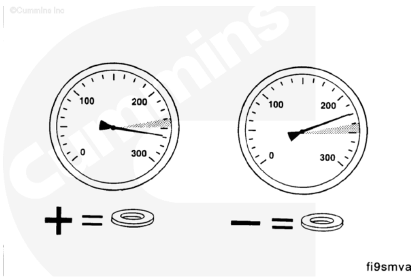
While testing the injectors, keep hands and body parts away from the injector nozzle. Fuel coming from the injector is under extreme pressure and can cause serious injury by penetrating the skin.

Use an injector nozzle tester, Part Number 3376946. All nozzles **must** be tested for opening pressure, chatter, and spray pattern.



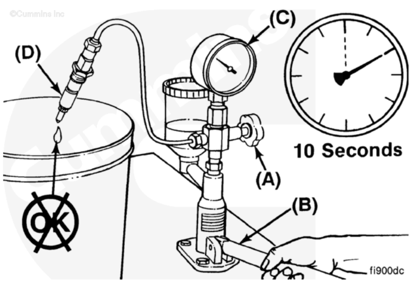
Check the injector opening pressure.

1. Open the valve.
2. Operate the lever at one stroke every second.
3. Read the pressure indicated when the injector spray begins.



If the opening pressure does **not** meet specifications, attempt one of the following solutions:

1. Add shims to increase pressure.
2. Remove shims to decrease pressure.

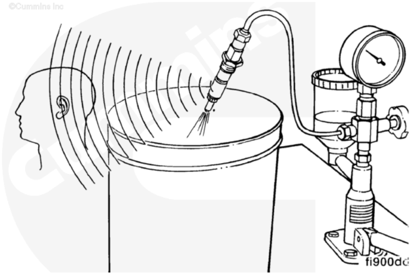


Leakage Test

1. Open the valve (A).
2. Operate the lever (B) to maintain a pressure of 20 bar [290 psi] below opening pressure (C).
3. No drops should fall from the tip (D) within 10 seconds.

Chatter Test

The chatter test indicates the ability of the needle valve to move freely and atomize the fuel correctly. An audible sound will possibly be heard as the valve rapidly opens and closes. A well-optimized spray pattern can possibly be seen.

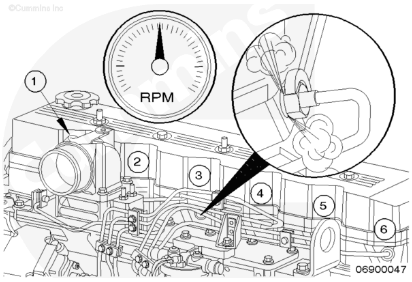


Note : Used nozzles must not be evaluated for chatter at lower speeds. A used nozzle can usually be used if it passes the leakage test.

Rear Gear Train

⚠ WARNING ⚠

Do not vent the fuel system on a hot engine; this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.



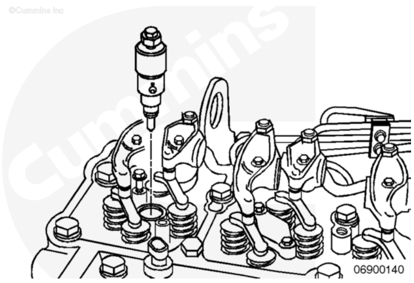
Test to determine which cylinder is misfiring, operate the engine, loosen the fuel line nut at one injector, and listen for a change in engine speed.

Note : A drop in engine speed indicates the injector was delivering fuel to the cylinder.

Note : Be sure to tighten the fuel line nut before proceeding to the next injector.

Check each cylinder until the malfunctioning injector is found.

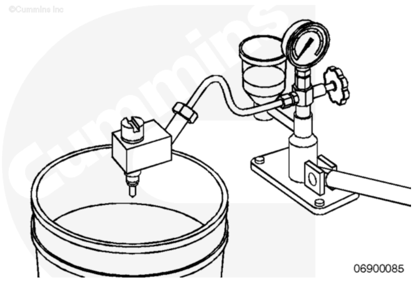
Remove the malfunctioning injector to test or replace it.



If the engine continues to misfire after replacing the injector, check for leaks in the high-pressure line.

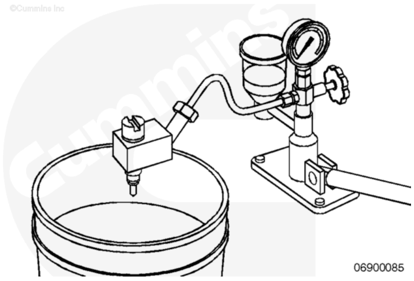
Check for a defective delivery valve that allows the fuel to drain back into the injection pump.

Use an injector holder, Part Number 3162269, to install the injector on the injector test stand, Part Number 3376946.



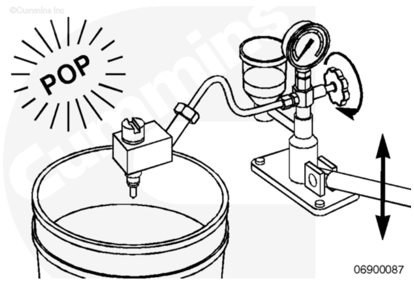
Open the bypass valve for the pressure gauge so the spray pattern can be checked.

Operate the test stand lever quickly several times to check the spray pattern of the injectors. Verify that the correct number of plumes are present for the number of holes the injector has. Also, pay close attention to the size and shape of each plume. If possible, compare the spray pattern to that of a new injector with the same assembly number.



Note : The injector spray pattern is an excellent indicator of the nozzle hole condition. Check each plume carefully; it is possible that only a single hole has damage. Significant performance problems will result if there is damage to any number of the holes.

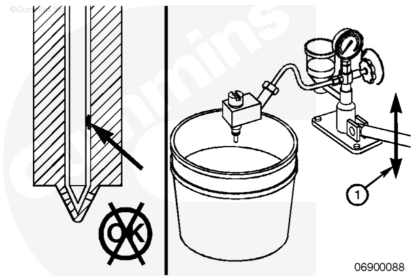
Close the bypass valve for the pressure gauge, and operate the test stand lever to check nozzle opening pressure.



There is a good crisp "pop" when the nozzle opens. The pressure specification is 300 ± 10 Bar [4351 ± 145 psi].

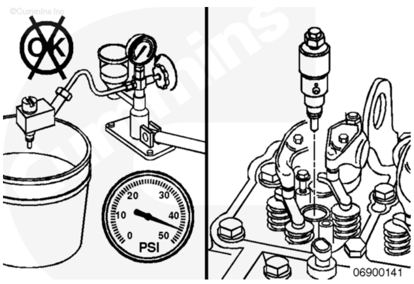
Note : If the nozzle opening pressure is out of specification, it is possible to add or remove shims to the injector to modify the opening pressure.

If the nozzle opening pressure is excessively low and/or the nozzle sprays excessive fuel, the injector needle is possibly sticking. The needle can be stuck because of poor lubrication or debris.



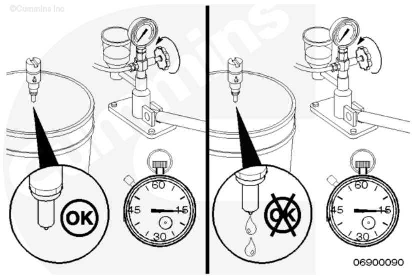
It is possible to unstick an injector needle by use of the injector test stand. Open the bypass valve for the pressure gauge and operate the test stand lever rapidly for 10 to 20 strokes.

Check the nozzle opening pressure and spray pattern again to determine if the injector has returned to normal operation.



If the injector remains out of specification, replace the injector.

Inspect the injector for dripping and/or excessive leak down.

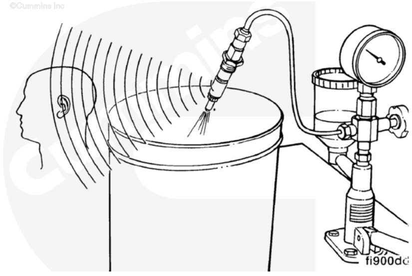


Close the bypass valve for the pressure gauge and build pressure to within 1000 kPa [145 psi] of the opening pressure of the nozzle.

Chatter Test

The chatter test indicates the ability of the needle valve to move freely and atomize the fuel correctly. An audible sound can be heard as the valve rapidly opens and closes. A well-optimized spray pattern can also be observed.

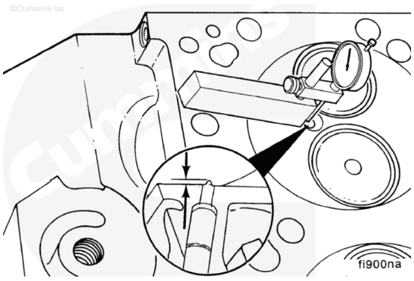
Note : Used nozzles must not be evaluated for chatter at lower speeds. A used nozzle can usually be used if it passes the leakage test.



Measure

⚠ CAUTION ⚠

The incorrect sealing washer can cause high-pressure fuel leaks and/or performance problems because of incorrect injector protrusion.



Install the depth gauge assembly, Tool Number 3164438, on the cylinder head combustion deck and zero it.

Rotate the depth gauge so that it measures the injector protrusion at the highest point on the injector.

Record the injector protrusion for each injector.

Injector Tip Protrusion B Series 1991 Automotive and CPL 1577		
mm		in
3.5	MIN	0.1378
4.5	MAX	0.1772

Injector Tip Protrusion All Other B Series Pre 1991 Automotive and 1994 Automotive		
mm		in
4.5	MIN	0.1772
5.5	MAX	0.2165

Injector Tip Protrusion Non-Automotive B Series		
mm		in
3.0	MIN	0.1180
4.0	MAX	0.1575

If the injector protrusion is not within the specifications, reference the injector seal thickness chart below for the available seal thicknesses.

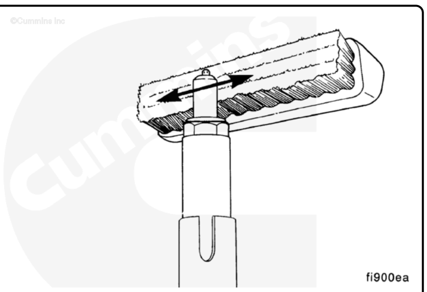
Injector Seal Thickness			
	mm		in
1.5 mm Seal	1.40	MIN	0.055
	1.68	MAX	0.066

	mm		in
2.5 mm Seal	2.40	MIN	0.095
	2.68	MAX	0.106

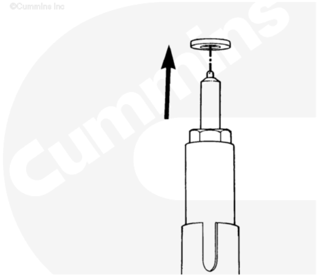
	mm		in
3.0 mm Seal	2.90	MIN	0.114
	3.18	MAX	0.125

Disassemble

Clean the carbon residue from the injector nozzle. Use a brass wire brush and a piece of hardwood dipped in test oil.



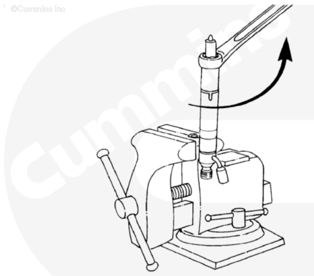
Remove the injector seal and discard the seal.



fi9wawa

Clamp an injector hold-down clamp in a soft-jawed vise to hold the injector.

Remove the injector nozzle retaining nut.

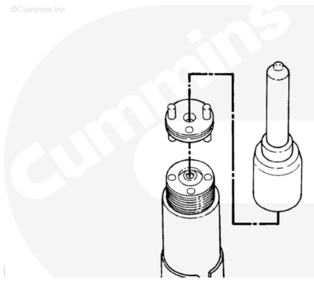


fi9numb

⚠ CAUTION ⚠

Place the injector nozzle and needle valve in a suitable bath of clean test oil or damage will occur.

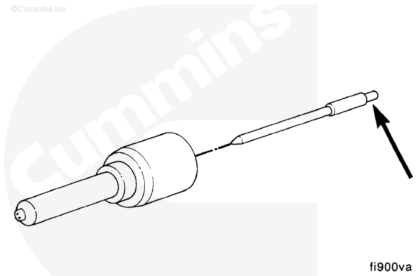
Remove the nozzle needle valve and intermediate plate.



fi9vama

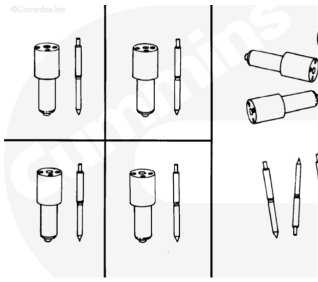
⚠ CAUTION ⚠

Hold the needle valve by the stem only. Contact from the skin will corrode the finely lapped surface.



fi900va

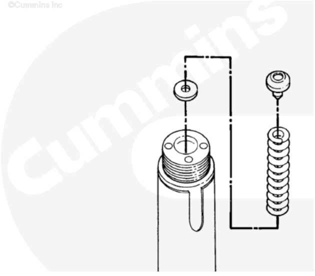
Note : The needle valve and nozzle tip are precisely matched for fit. The parts must not be intermixed.



fi900wa

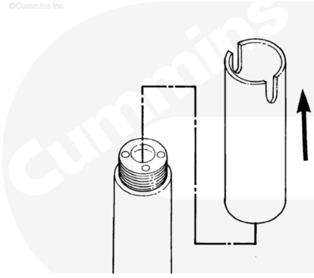
Remove the nozzle holder from the vise.

Remove the pressure spindle, pressure spring, and shims.



fi900fa

Remove and discard the injector sealing sleeve.

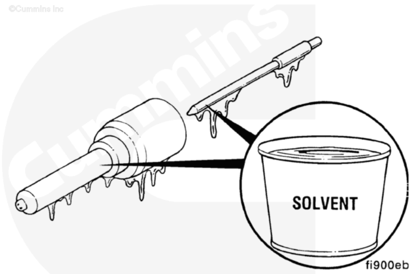


fi9slma

Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

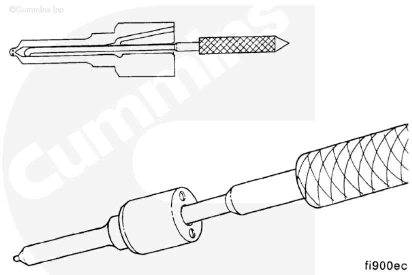


fi900eb

Rinse the nozzle bodies and needle valves in solvent to flush them thoroughly and completely remove all varnish and carbon deposits.

⚠ CAUTION ⚠

Never use emery paper, a steel brush, or any other metal scraper to clean the nozzle. The parts can be damaged.



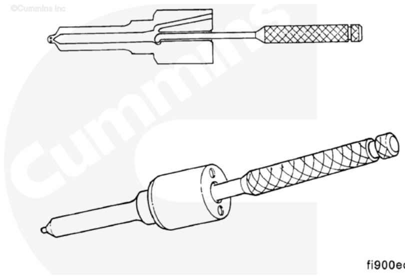
fi900ec

Dip the nozzle seat in clean test oil, and use the nozzle cleaning kit, Part Number 3376947, to clean the nozzle seat. Polish the needle seat with a piece of hardwood dipped in test oil.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

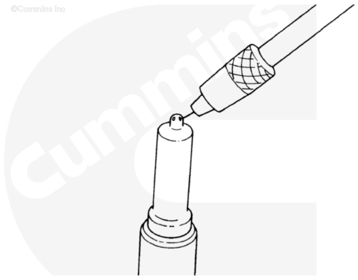
Clean the interior ring groove of the nozzle with a scraper, as illustrated. Rinse the nozzle in solvent to remove all dirt and carbon residue, and dip it in clean test oil.



fi900ed

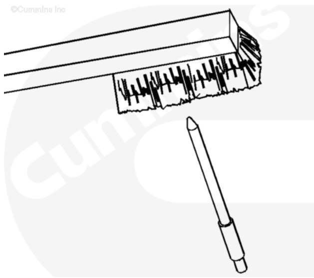
Clean the spray holes with an appropriate size cleaning needle, as illustrated.

Remove burned-on combustion deposits on all nozzles with a commercially available cleaner. Rinse all parts in clean test oil.



fi900ee

Clean the needle valve tip with a brass brush.

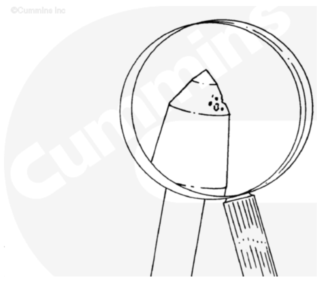


06900173

Inspect

Front Gear Train

Inspect the injector. Inspect the o-ring for damage. Inspect for burrs on the inlet to the injector. Check the nozzle holes for any signs of damage such as hole erosion or hole plugging. Also, check the nozzle color for signs of overheating. Overheating will cause the nozzle to turn a dark yellow/tan or blue color, depending on the temperature of the overheat.



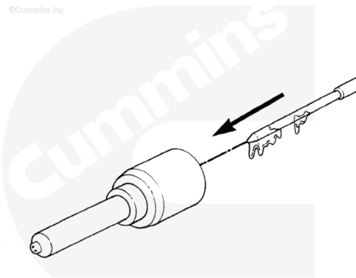
fi9vasa

Inspect for rough surfaces and erosion. The pressure shoulder will normally have a rough machined appearance.

Inspect the injector bore for old sealing washers.

Note : Deteriorated needle valves must be replaced as a matched unit with their compatible nozzle body.

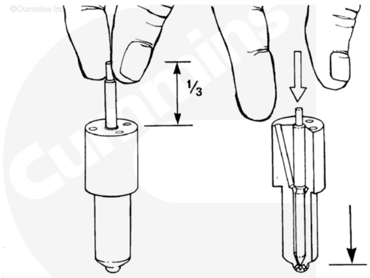
Dip the needle valve in clean test oil and insert the needle valve all the way into the nozzle body.


fi9vawa

Pull the needle valve one-third of the way out of the nozzle body. With the needle valve in the vertical position, the needle valve **must** slide all the way back into the nozzle body under its own weight.

If the nozzle fails the slide test, clean and test the nozzle again.

Note : Any needle valve and nozzle body assembly that does not pass this test must be replaced.

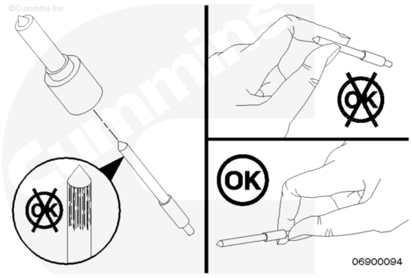


fi9vasb

Rear Gear Train

⚠ CAUTION ⚠

Do not touch the needle with your fingertips. Oil from your fingers will damage the needle. Hold the needle by the stem. If the needle is touched, wipe clean with a soft cloth and dip in clean diesel fuel.



06900094

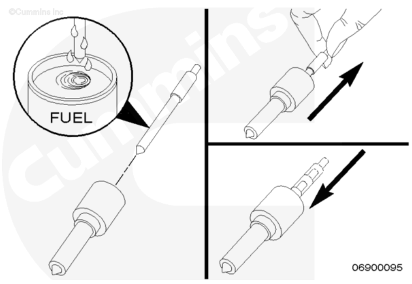
Remove the needle from the nozzle tip and inspect the color.

Check for signs of excessive carbon or overheating (dark yellow/tan or blue needle color).

Check for scuff marks on the needle.

Dip the needle in clean diesel fuel and insert it into the nozzle.

Hold the nozzle at a 45-degree angle and pull the needle 2/3 of the way out. Under its own weight, the needle **must** slide smoothly back into the nozzle.



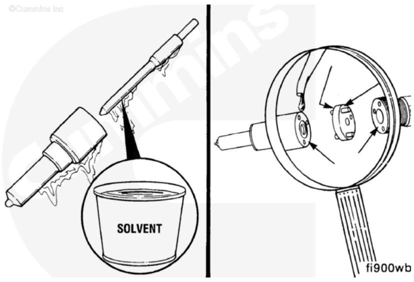
06900095

Assemble

Front Gear Train

⚠ WARNING ⚠

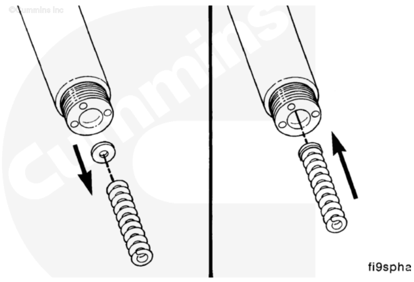
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



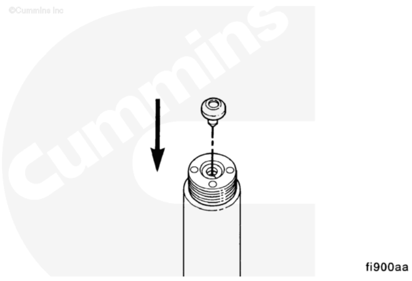
Note : Make sure that all mating surfaces and pressure faces are thoroughly cleaned and lubricated with test oil before assembly. New nozzles **must** be cleaned and lubricated before assembly.

Note : Install the same thickness of shims that were removed in disassembly. Use the pressure spring to make sure the shims are installed flat.

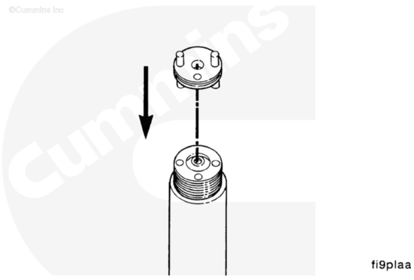
Install the shims and pressure spring.



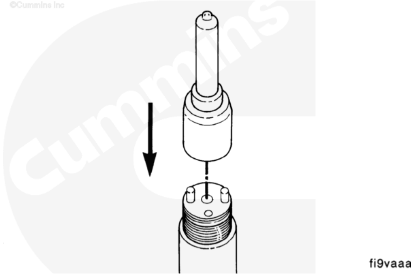
Install the spindle.



Install the intermediate plate.

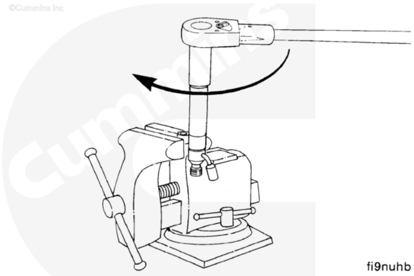


Install the needle valve and nozzle assembly.



Install the nozzle retaining nut.

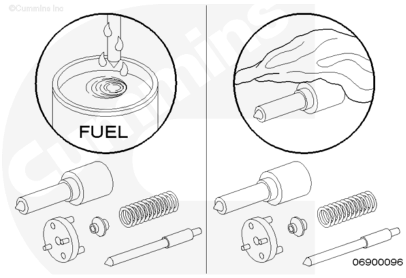
Torque Value: 30 n•m [22 ft-lb]



Rear Gear Train

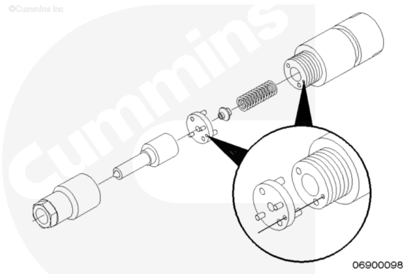
Clean the internal components of the injector with clean diesel fuel and a clean cloth.

Make sure there is **no** debris in the internal parts of the injector.



Note : Make sure the intermediate plate is in the correct orientation, with the supply hole on the plate lining up with the supply hole on the holder.

Install the spring, button, intermediate plate, and nozzle/needle.



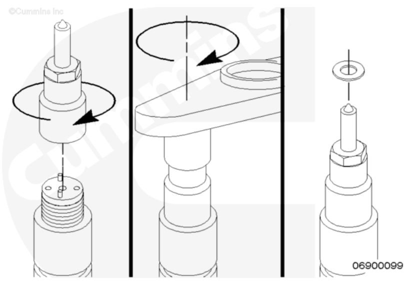
Install the retaining nut hand-tight.

Place the injector in the injector clamp.

Tighten the retaining nut.

Torque Value: 47 n•m [35 ft-lb]

Install the sealing washer.



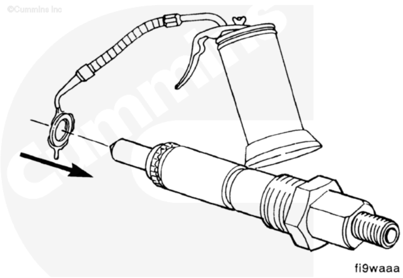
Install

Front Gear Train

Assemble the injector and a new copper sealing washer.

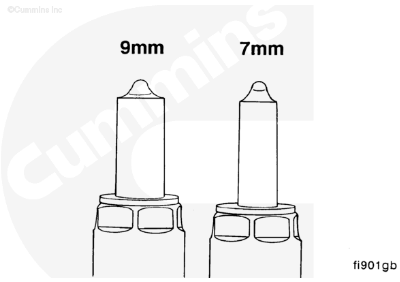
Use **only** one copper washer.

Service Tip: A light coat of clean lubricating engine oil between the washer and injector can help to keep the washer from falling during installation.

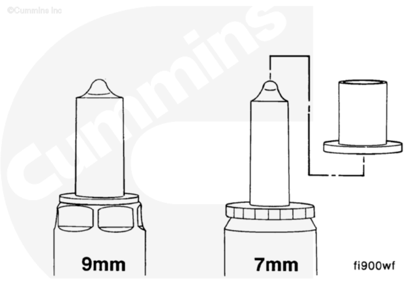


⚠ CAUTION ⚠

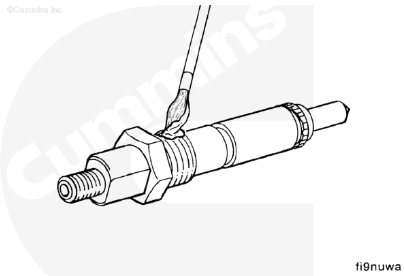
Early model injectors (pre-1991) have a 9-mm injector tip that can not be used in engines built in 1991 or later as these engines use a 7-mm injector tip.



If the special adapter sleeve is installed onto the 7-mm injector tip, 7-mm injectors can be used in early model (9-mm) injector holes.



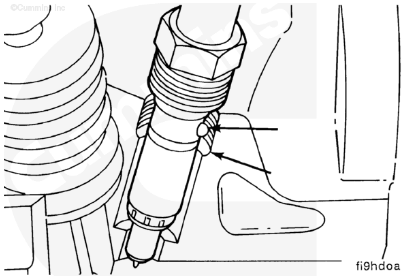
Apply a coat of anti-seize compound, Part Number 3824879, to the threads of the injector hold-down nut and between the top of the nut and the injector body.



Note : Align the injector's protrusion with the notch in the bore.

Note : The present Bosch® injector has an o-ring located above the hold-down nut. After tightening the injector, be sure to push the o-ring into the groove.

Torque Value: 60 n•m [44 ft-lb]

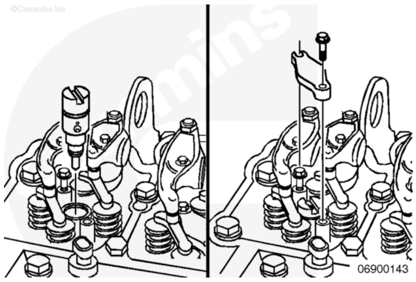


Rear Gear Train

Place the injector in the head in the proper orientation.

Install the injector hold-down and tighten.

Torque Value: 10 n•m [89 in-lb]



Finishing Steps

Front Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the high-pressure fuel supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)
- Install the fuel drain manifold. Refer to Procedure 006-021 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-021-tr.html)
- Connect the batteries.
- Bleed all air from the fuel system. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)
- Operate the engine and check for leaks.

Rear Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the fuel connector. Refer to Procedure 006-052 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-052-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Install the high-pressure fuel lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)
- Connect the batteries.
- Bleed all the air from the fuel system. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)
- Operate the engine and check for leaks.

Last Modified: 07-Feb-2012